

Math 420/620, Mathematical Modeling
Homework # 1

Homework due at the time and date in WebCampus.

Use Maple for this assignment.

Note: The five step method is to be used in part (a) but is not used in the subsequent parts.

A local daily newspaper with a circulation of 80,000 subscribers is thinking of raising its subscription price,. Currently the price is \$1.50 per week, and it is estimated that the paper would loose 5,000 subscribers if the rate were to be raised by 10 cents/week. Further, it is assumed that s , the number of subscribers, is a linear function of the subscription price.

a) Find the subscription price that maximizes profit. Use the five-step method, and model as a one-variable optimization problem.

Notes: Let x be the subscription price (\$/paper); s be the number of subscriptions (papers); y be profit (\$). The profit function for y should be a quadratic function in x .

b) Let $n=5,000$ denote the number of subscribers lost when the subscription price increases by ten cents.

Calculate the optimal subscription price x as a function of n , and use this formula to determine the sensitivity $S(x, n)$.

c) Use the sensitivity $S(x, n)$ found in part (b) to predict

- (i) the effect of a 20% **increase** in $n = 5000$ on the optimal subscription price x ; In other words, what does the sensitivity give as the new optimal subscription price x ?
- (ii) the effect of a 20% **decrease** in $n = 5000$ on the optimal subscription price x . In other words, what does the sensitivity give as the new optimal subscription price x ?